

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418604
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Kim; Yongwon

Electronic apparatus, electronic apparatus case, and method for manufacturing electronic apparatus case

Abstract

An electronic apparatus according to various embodiments disclosed in the present document includes a frame, a plate arranged on an inner side of the frame, an antenna member including a plurality of radiating units that transmit or receive an RF signal, a receiving space provided between the frame and the plate so that the antenna member is arranged between the frame and the plate and a plurality of joint parts, which are parts at which the frame and the plate are joined. A reinforcing joint part included in the plurality of joint parts may be positioned between adjacent radiating units from among the plurality of radiating units of the antenna member. Various other embodiments may be possible.

Inventors:	Kim; Yongwon (Suwon-si, KR)
Applicant:	Samsung Electronics Co., Ltd. (Suwon-si, KR)
Family ID:	1000008822170
Assignee:	SAMSUNG ELECTRONICS CO., LTD. (Gyeonggi-Do, KR)
Appl. No.:	18/186193
Filed:	March 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230247122 A1	Aug. 03, 2023

Foreign Application Priority Data

KR	10-2020-0122195	Sep. 22, 2020
KR	10-2021-0035755	Mar. 19, 2021

Related U.S. Application Data

continuation parent-doc WO PCT/KR2021/012906 20210917 PENDING child-doc US 18186193

Publication Classification

Int. Cl.: **H01Q1/24** (20060101); **G06F1/16** (20060101); **H04M1/02** (20060101); **H05K9/00** (20060101)

U.S. Cl.:

CPC **H04M1/0266** (20130101); **G06F1/1628** (20130101); **H01Q1/24** (20130101); **H05K9/0024** (20130101);

Field of Classification Search

CPC: G06F (1/16); G06F (1/1628); G06F (1/1626); H01Q (1/24); H01Q (1/243); H01Q (1/38); H01Q (21/08); H04M (1/02); H04M (1/0216); H04M (1/0266); H05K (9/00); H05K (9/0024); B23K (26/21)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5256856	12/1992	Takano et al.	N/A	N/A
5497937	12/1995	Yoshikawa et al.	N/A	N/A
6337144	12/2001	Shimizu et al.	N/A	N/A
10201102	12/2018	Lee	N/A	H05K 5/0217
10567561	12/2019	Kang	N/A	H04M 1/18
10665926	12/2019	Kim et al.	N/A	N/A
10691168	12/2019	Huang	N/A	N/A
10908352	12/2020	Chen	N/A	G02B 6/0088
12047522	12/2023	Kim	N/A	H05K 1/14
12176620	12/2023	Lee	N/A	H01Q 9/045
2002/0151328	12/2001	Shin et al.	N/A	N/A
2003/0085211	12/2002	Ito et al.	N/A	N/A
2013/0320098	12/2012	Mukai et al.	N/A	N/A
2019/0070760	12/2018	Huang et al.	N/A	N/A
2020/0076057	12/2019	Leutheuser et al.	N/A	N/A
2020/0260605	12/2019	Lee et al.	N/A	N/A
2021/0273322	12/2020	Kim et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
5344108	12/2012	JP	N/A
920019463	12/1991	KR	N/A
950005430	12/1994	KR	N/A
950018656	12/1994	KR	N/A
0115493	12/1997	KR	N/A

19990045200	12/1998	KR	N/A
20010057540	12/2000	KR	N/A
20030017424	12/2002	KR	N/A
200369702	12/2003	KR	N/A
20060025359	12/2005	KR	N/A
20060074295	12/2005	KR	N/A
100699408	12/2006	KR	N/A
20130005164	12/2012	KR	N/A
20130055910	12/2012	KR	N/A
20160099264	12/2015	KR	N/A
20170037464	12/2016	KR	N/A
101775004	12/2016	KR	N/A
101795491	12/2016	KR	N/A
20190037739	12/2018	KR	N/A
102079822	12/2019	KR	N/A
20200034349	12/2019	KR	N/A
20200063186	12/2019	KR	N/A

OTHER PUBLICATIONS

Search Report issued in European Patent Application No. 21872884.8; Application Filing Date Sep. 17, 2021; Date of Mailing Feb. 6, 2024 (9 pages). cited by applicant
International Search Report with Written Opinion for International Application No. PCT/KR2021/012906; Application Filing Date Sep. 17, 2021; Date of Mailing Jan. 3, 2022 (15 pages). cited by applicant

Primary Examiner: Phan; Tho G

Attorney, Agent or Firm: CANTOR COLBURN LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, claiming priority under § 365(c), of an International application No. PCT/KR2021/012906, filed on Sep. 17, 2021, which is based on and claims the benefit of a Korean patent application number 10-2021-0035755, filed on Mar. 19, 2021, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2020-0122195, filed on Sep. 22, 2020, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) Various embodiments disclosed in the present document relate to an electronic apparatus, an electronic apparatus case, and a method of manufacturing an electronic apparatus case.

BACKGROUND ART

(2) Cases, which define external appearances of electronic apparatuses, are manufactured by various methods. Because a case is an important element that defines an aesthetic external appearance of the electronic apparatus, the case, to which various design elements are added, is also manufactured.

(3) The case is a constituent element that supports various electronic components in the electronic apparatus. One of the important design elements is to design the case so that the case may protect

various electronic components disposed in the electronic apparatus.

DISCLOSURE OF INVENTION

Technical Problem

(4) The number of functions and the number of electronic components included in the electronic apparatus have been increased. However, because the size of the electronic apparatus gradually decreases, a space in which the electronic components need to be disposed becomes insufficient. In addition, a weight of a portable electronic apparatus is preferably decreased to provide convenience and usability for a user.

(5) Under this circumstance, it is necessary to ensure rigidity of the case that defines an external appearance of the electronic apparatus.

(6) Further, with the development of communication technologies, there is also a need to dispose a particular constituent element for performing communication functions so that the particular constituent element is disposed adjacent to an electronic apparatus case.

(7) Various embodiments disclosed in the present document may provide a structure capable of ensuring rigidity of a case while disposing a constituent element adjacent to a side surface of the case for performing a communication function.

Solution to Problem

(8) Various embodiments disclosed in the present document provide an electronic apparatus including: a frame; a plate disposed in the frame; an antenna member including a plurality of radiation parts configured to transmit or receive an RF signal; an accommodation space provided between the frame and the plate so that the antenna member is disposed between the frame and the plate; and a plurality of junction parts that is parts by which the frame and the plate are joined, in which a reinforcement junction part included in the plurality of junction parts is positioned between the adjacent radiation parts among the plurality of radiation parts of the antenna member.

(9) Various embodiments disclosed in the present document provide an electronic apparatus case including: a frame; a plate disposed in the frame; a plurality of junction parts that is parts by which the frame and the plate are joined; and an intermediate member formed between the frame and the plate, in which a reinforcement junction part included in the plurality of junction parts is positioned between adjacent radiation parts among a plurality of radiation parts included in an antenna member disposed in an accommodation space provided between the frame and the plate.

(10) Various embodiments disclosed in the present document provide a method of manufacturing an electronic apparatus case, the method including: forming a plate and a frame; temporarily assembling the frame to the plate so that a junction protrusion formed on one of the frame and the plate is inserted into a junction groove formed in the other of the frame and the plate; joining the junction protrusion and the junction groove; forming an intermediate member in a space between the frame and the plate; and cutting at least one of the frame, the plate, and the intermediate member, in which the cutting of the at least one of the frame, the plate, and the intermediate member includes forming an accommodation space in which an antenna member is disposed between the frame and the plate, and in which a reinforcement junction protrusion included in the junction protrusion and a reinforcement junction groove included in the junction groove is positioned between adjacent radiation parts among a plurality of radiation parts included in the antenna member disposed in the accommodation space.

Advantageous Effects of Invention

(11) According to various embodiments disclosed in the present document, the antenna may be disposed on the side surface of the case so that the communication signal may be smoothly transmitted or received. In addition, it is possible to improve overall durability of the electronic apparatus case by ensuring rigidity of the portion where the antenna is disposed.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) In connection with the description of the drawings, the same or similar reference numerals may be used for the same or similar components.
- (2) FIG. 1 is a block diagram of an electronic device in a network environment according to various embodiments.
- (3) FIG. 2 is a block diagram of the electronic device for supporting legacy network communication and 5G network communication according to various embodiments.
- (4) FIG. 3A is a perspective view of a front surface of an electronic apparatus according to various embodiments disclosed in the present document.
- (5) FIG. 3B is a perspective view of a rear surface of the electronic apparatus in FIG. 3A according to various embodiments disclosed in the present document.
- (6) FIG. 3C is a deployed perspective view of the electronic apparatus in FIG. 3A according to various embodiments disclosed in the present document.
- (7) FIG. 4A(a), (b), (c) are views illustrating an embodiment of a structure of a third antenna module **246** illustrated in FIG. 2.
- (8) FIG. 4B is a cross-sectional view taken along line Y-Y' of the third antenna module **246** illustrated in FIG. 4A(a).
- (9) FIG. 5A is an exploded perspective view of an electronic apparatus case according to various embodiments disclosed in the present document.
- (10) FIG. 5B is a front view of the electronic apparatus case according to various embodiments disclosed in the present document.
- (11) FIG. 6 is a perspective view illustrating an antenna member and the electronic apparatus case according to various embodiments disclosed in the present document.
- (12) FIG. 7 is a perspective view of the antenna member according to various embodiments disclosed in the present document.
- (13) FIG. 8A is a view for explaining coupling between a plate and a frame according to various embodiments disclosed in the present document.
- (14) FIG. 8B is an enlarged view of area P illustrated in FIG. 8A.
- (15) FIG. 9 is a view for explaining a positional relationship between the antenna member and a reinforcement junction part according to various embodiments disclosed in the present document.
- (16) FIG. 10A is a flowchart illustrating a method of manufacturing the electronic apparatus case according to various embodiments disclosed in the present document.
- (17) FIG. 10B(a), (b), (c), (d) is a view illustrating a state of an electronic apparatus case manufactured according to various embodiments disclosed in the present document.
- (18) FIG. 10C is a view illustrating states of an accommodation space and a peripheral portion of the electronic apparatus case manufactured according to various embodiments disclosed in the present document.
- (19) FIG. 11(a), (b) is a cross-sectional view taken by cutting the electronic apparatus case illustrated in FIG. 10C (d).

MODE FOR THE INVENTION

- (20) FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connection terminal **178**, a

haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connection terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(21) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(22) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(23) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(24) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(25) The input module **150** may receive a command or data to be used by another component (e.g.,

the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(26) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(27) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(28) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(29) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(30) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(31) A connection terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connection terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(32) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(33) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(34) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(35) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(36) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101**

and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(37) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(38) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(39) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit

board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(40) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(41) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(42) FIG. 2 is a block diagram illustrating an electronic device in a network environment **200** including a plurality of cellular networks according to various embodiments of the disclosure.

(43) Referring to FIG. 2, the electronic device **101** may include a first communication processor **212**, second communication processor **214**, first RFIC **222**, second RFIC **224**, third RFIC **226**, fourth RFIC **228**, first radio frequency front end (RFFE) **232**, second RFFE **234**, first antenna module **242**, second antenna module **244**, and antenna **248**. The electronic device **101** may include a processor **120** and a memory **130**. A second network **199** may include a first cellular network **292** and a second cellular network **294**. According to another embodiment, the electronic device **101** may further include at least one of the components described with reference to FIG. 1, and the second network **199** may further include at least one other network. According to one embodiment, the first communication processor **212**, second communication processor **214**, first RFIC **222**, second RFIC **224**, fourth RFIC **228**, first RFFE **232**, and second RFFE **234** may form at least part of the wireless communication module **192**. According to another embodiment, the fourth RFIC **228** may be omitted or included as part of the third RFIC **226**.

(44) The first communication processor **212** may establish a communication channel of a band to be used for wireless communication with the first cellular network **292** and support legacy network communication through the established communication channel. According to various embodiments, the first cellular network may be a legacy network including a second generation (2G), 3G, 4G, or long term evolution (LTE) network. The second communication processor **214**

may establish a communication channel corresponding to a designated band (e.g., about 6 GHz to about 60 GHz) of bands to be used for wireless communication with the second cellular network **294**, and support 5G network communication through the established communication channel. According to various embodiments, the second cellular network **294** may be a 5G network defined in 3GPP. Additionally, according to an embodiment, the first communication processor **212** or the second communication processor **214** may establish a communication channel corresponding to another designated band (e.g., about 6 GHz or less) of bands to be used for wireless communication with the second cellular network **294** and support 5G network communication through the established communication channel. According to one embodiment, the first communication processor **212** and the second communication processor **214** may be implemented in a single chip or a single package. According to various embodiments, the first communication processor **212** or the second communication processor **214** may be formed in a single chip or a single package with the processor **120**, the auxiliary processor **123**, or the communication module **190**.

(45) Upon transmission, the first RFIC **222** may convert a baseband signal generated by the first communication processor **212** to a radio frequency (RF) signal of about 700 MHz to about 3 GHz used in the first cellular network **292** (e.g., legacy network). Upon reception, an RF signal may be obtained from the first cellular network **292** (e.g., legacy network) through an antenna (e.g., the first antenna module **242**) and be preprocessed through an RFFE (e.g., the first RFFE **232**). The first RFIC **222** may convert the preprocessed RF signal to a baseband signal so as to be processed by the first communication processor **212**.

(46) Upon transmission, the second RFIC **224** may convert a baseband signal generated by the first communication processor **212** or the second communication processor **214** to an RF signal (hereinafter, 5G Sub6 RF signal) of a Sub6 band (e.g., 6 GHz or less) to be used in the second cellular network **294** (e.g., 5G network). Upon reception, a 5G Sub6 RF signal may be obtained from the second cellular network **294** (e.g., 5G network) through an antenna (e.g., the second antenna module **244**) and be pretreated through an RFFE (e.g., the second RFFE **234**). The second RFIC **224** may convert the preprocessed 5G Sub6 RF signal to a baseband signal so as to be processed by a corresponding communication processor of the first communication processor **212** or the second communication processor **214**.

(47) The third RFIC **226** may convert a baseband signal generated by the second communication processor **214** to an RF signal (hereinafter, 5G Above6 RF signal) of a 5G Above6 band (e.g., about 6 GHz to about 60 GHz) to be used in the second cellular network **294** (e.g., 5G network). Upon reception, a 5G Above6 RF signal may be obtained from the second cellular network **294** (e.g., 5G network) through an antenna (e.g., the antenna **248**) and be preprocessed through the third RFFE **236**. The third RFIC **226** may convert the preprocessed 5G Above6 RF signal to a baseband signal so as to be processed by the second communication processor **214**. According to one embodiment, the third RFFE **236** may be formed as part of the third RFIC **226**.

(48) According to an embodiment, the electronic device **101** may include a fourth RFIC **228** separately from the third RFIC **226** or as at least part of the third RFIC **226**. In this case, the fourth RFIC **228** may convert a baseband signal generated by the second communication processor **214** to an RF signal (hereinafter, an intermediate frequency (IF) signal) of an intermediate frequency band (e.g., about 9 GHz to about 11 GHz) and transfer the IF signal to the third RFIC **226**. The third RFIC **226** may convert the IF signal to a 5G Above 6RF signal. Upon reception, the 5G Above 6RF signal may be received from the second cellular network **294** (e.g., a 5G network) through an antenna (e.g., the antenna **248**) and be converted to an IF signal by the third RFIC **226**. The fourth RFIC **228** may convert an IF signal to a baseband signal so as to be processed by the second communication processor **214**.

(49) According to one embodiment, the first RFIC **222** and the second RFIC **224** may be implemented into at least part of a single package or a single chip. According to one embodiment, the first RFFE **232** and the second RFFE **234** may be implemented into at least part of a single

package or a single chip. According to one embodiment, at least one of the first antenna module **242** or the second antenna module **244** may be omitted or may be combined with another antenna module to process RF signals of a corresponding plurality of bands.

(50) According to one embodiment, the third RFIC **226** and the antenna **248** may be disposed at the same substrate to form a third antenna module **246**. For example, the wireless communication module **192** or the processor **120** may be disposed at a first substrate (e.g., main PCB). In this case, the third RFIC **226** is disposed in a partial area (e.g., lower surface) of the first substrate and a separate second substrate (e.g., sub PCB), and the antenna **248** is disposed in another partial area (e.g., upper surface) thereof; thus, the third antenna module **246** may be formed. By disposing the third RFIC **226** and the antenna **248** in the same substrate, a length of a transmission line therebetween can be reduced. This may reduce, for example, a loss (e.g., attenuation) of a signal of a high frequency band (e.g., about 6 GHz to about 60 GHz) to be used in 5G network communication by a transmission line. Therefore, the electronic device **101** may improve a quality or speed of communication with the second cellular network **294** (e.g., 5G network).

(51) According to one embodiment, the antenna **248** may be formed in an antenna array including a plurality of antenna elements that may be used for beamforming. In this case, the third RFIC **226** may include a plurality of phase shifters **238** corresponding to a plurality of antenna elements, for example, as part of the third RFFE **236**. Upon transmission, each of the plurality of phase shifters **238** may convert a phase of a 5G Above6 RF signal to be transmitted to the outside (e.g., a base station of a 5G network) of the electronic device **101** through a corresponding antenna element. Upon reception, each of the plurality of phase shifters **238** may convert a phase of the 5G Above6 RF signal received from the outside to the same phase or substantially the same phase through a corresponding antenna element. This enables transmission or reception through beamforming between the electronic device **101** and the outside.

(52) The second cellular network **294** (e.g., 5G network) may operate (e.g., stand-alone (SA)) independently of the first cellular network **292** (e.g., legacy network) or may be operated (e.g., non-stand alone (NSA)) in connection with the first cellular network **292**. For example, the 5G network may have only an access network (e.g., 5G radio access network (RAN) or a next generation (NG) RAN and have no core network (e.g., next generation core (NGC)). In this case, after accessing to the access network of the 5G network, the electronic device **101** may access to an external network (e.g., Internet) under the control of a core network (e.g., an evolved packed core (EPC)) of the legacy network. Protocol information (e.g., LTE protocol information) for communication with a legacy network or protocol information (e.g., new radio (NR) protocol information) for communication with a 5G network may be stored in the memory **130** to be accessed by other components (e.g., the processor **120**, the first communication processor **212**, or the second communication processor **214**).

(53) FIG. 3A illustrates a perspective view showing a front surface of a mobile electronic device **300** according to an embodiment, and FIG. 3B illustrates a perspective view showing a rear surface of the mobile electronic device **300** shown in FIG. 3A.

(54) The electronic device **300** of FIGS. 3A and 3B may be at least partially similar to the electronic device **101** of FIG. 1 or may include other embodiments of the electronic device.

(55) Referring to FIGS. 3A and 3B, the mobile electronic device **300** may include a housing **310** that includes a first surface (or front surface) **310A**, a second surface (or rear surface) **310B**, and a lateral surface **310C** that surrounds a space between the first surface **310A** and the second surface **310B**. The housing **310** may refer to a structure that forms a part of the first surface **310A**, the second surface **310B**, and the lateral surface **310C**. The first surface **310A** may be formed of a front plate **302** (e.g., a glass plate or polymer plate coated with a variety of coating layers) at least a part of which is substantially transparent. The second surface **310B** may be formed of a rear plate **311** which is substantially opaque. The rear plate **311** may be formed of, for example, coated or colored glass, ceramic, polymer, metal (e.g., aluminum, stainless steel (STS), or magnesium), or any

combination thereof. The lateral surface **310C** may be formed of a lateral bezel structure (or “lateral member”) **320** which is combined with the front plate **302** and the rear plate **311** and includes a metal and/or polymer. The rear plate **311** and the lateral bezel structure **320** may be integrally formed and may be of the same material (e.g., a metallic material such as aluminum).

(56) The front plate **302** may include two first regions **310D** disposed at long edges thereof, respectively, and bent and extended seamlessly from the first surface **310A** toward the rear plate **311**. Similarly, the rear plate **311** may include two second regions **310E** disposed at long edges thereof, respectively, and bent and extended seamlessly from the second surface **310B** toward the front plate **302**. The front plate **302** (or the rear plate **311**) may include only one of the first regions **310D** (or of the second regions **310E**). The first regions **310D** or the second regions **310E** may be omitted in part. When viewed from a lateral side of the mobile electronic device **300**, the lateral bezel structure **320** may have a first thickness (or width) on a lateral side where the first region **310D** or the second region **310E** is not included, and may have a second thickness, being less than the first thickness, on another lateral side where the first region **310D** or the second region **310E** is included.

(57) The mobile electronic device **300** may include at least one of a display **301**, audio modules **303**, **307** and **314**, sensor modules **304** and **319**, camera modules **305**, **312** and **313**, a key input device **317**, a light emitting device, and connector holes **308** and **309**. The mobile electronic device **300** may omit at least one (e.g., the key input device **317** or the light emitting device) of the above components, or may further include other components.

(58) The display **301** may be exposed through a substantial portion of the front plate **302**, for example. At least a part of the display **301** may be exposed through the front plate **302** that forms the first surface **310A** and the first region **310D** of the lateral surface **310C**. Outlines (i.e., edges and corners) of the display **301** may have substantially the same form as those of the front plate **302**. The spacing between the outline of the display **301** and the outline of the front plate **302** may be substantially unchanged in order to enlarge the exposed area of the display **301**. A recess or opening may be formed in a portion of a display area of the display **301** to accommodate at least one of the audio module **314**, the sensor module **304**, the camera module **305**, and the light emitting device. At least one of the audio module **314**, the sensor module **304**, the camera module **305**, the fingerprint sensor (not shown), and the light emitting element may be disposed on the back of the display area of the display **301**. The display **301** may be combined with, or adjacent to, a touch sensing circuit, a pressure sensor capable of measuring the touch strength (pressure), and/or a digitizer for detecting a stylus pen. At least a part of the sensor modules **304** and **319** and/or at least a part of the key input device **317** may be disposed in the first region **310D** and/or the second region **310E**.

(59) The audio modules **303**, **307** and **314** may correspond to a microphone hole **303** and speaker holes **307** and **314**, respectively. The microphone hole **303** may contain a microphone disposed therein for acquiring external sounds and, in a case, contain a plurality of microphones to sense a sound direction. The speaker holes **307** and **314** may be classified into an external speaker hole **307** and a call receiver hole **314**. The microphone hole **303** and the speaker holes **307** and **314** may be implemented as a single hole, or a speaker (e.g., a piezo speaker) may be provided without the speaker holes **307** and **314**.

(60) The sensor modules **304** and **319** may generate electrical signals or data corresponding to an internal operating state of the mobile electronic device **300** or to an external environmental condition. The sensor modules **304** and **319** may include a first sensor module **304** (e.g., a proximity sensor) and/or a second sensor module (e.g., a fingerprint sensor) disposed on the first surface **310A** of the housing **310**, and/or a third sensor module **319** (e.g., a heart rate monitor (HRM) sensor) and/or a fourth sensor module (e.g., a fingerprint sensor) disposed on the second surface **310B** of the housing **310**. The fingerprint sensor may be disposed on the second surface **310B** as well as the first surface **310A** (e.g., the display **301**) of the housing **310**. The electronic

device **300** may further include at least one of a gesture sensor, a gyro sensor, an air pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(61) The camera modules **305**, **312** and **313** may include a first camera device **305** disposed on the first surface **310A** of the electronic device **300**, and a second camera device **312** and/or a flash **313** disposed on the second surface **310B**. The camera module **305** or the camera module **312** may include one or more lenses, an image sensor, and/or an image signal processor. The flash **313** may include, for example, a light emitting diode or a xenon lamp. Two or more lenses (infrared cameras, wide angle and telephoto lenses) and image sensors may be disposed on one side of the electronic device **300**.

(62) The key input device **317** may be disposed on the lateral surface **310C** of the housing **310**. The mobile electronic device **300** may not include some or all of the key input device **317** described above, and the key input device **317** which is not included may be implemented in another form such as a soft key on the display **301**. The key input device **317** may include the sensor module disposed on the second surface **310B** of the housing **310**.

(63) The light emitting device may be disposed on the first surface **310A** of the housing **310**. For example, the light emitting device may provide status information of the electronic device **300** in an optical form. The light emitting device may provide a light source associated with the operation of the camera module **305**. The light emitting device may include, for example, a light emitting diode (LED), an IR LED, or a xenon lamp.

(64) The connector holes **308** and **309** may include a first connector hole **308** adapted for a connector (e.g., a USB connector) for transmitting and receiving power and/or data to and from an external electronic device, and/or a second connector hole **309** adapted for a connector (e.g., an earphone jack) for transmitting and receiving an audio signal to and from an external electronic device.

(65) Some sensor modules **305** of camera modules **305** and **312**, some sensor modules **304** of sensor modules **304** and **319**, or an indicator may be arranged to be exposed through a display **301**. For example, the camera module **305**, the sensor module **304**, or the indicator may be arranged in the internal space of an electronic device **300** so as to be brought into contact with an external environment through an opening of the display **301**, which is perforated up to a front plate **302**. In another embodiment, some sensor modules **304** may be arranged to perform their functions without being visually exposed through the front plate **302** in the internal space of the electronic device. For example, in this case, an area of the display **301** facing the sensor module may not require a perforated opening.

(66) FIG. 3C illustrates an exploded perspective view showing a mobile electronic device **300** shown in FIG. 3A.

(67) Referring to FIG. 3C, the mobile electronic device **300** may include a lateral bezel structure **310**, a first support member **3211** (e.g., a bracket), a front plate **302**, a display **301**, an electromagnetic induction panel, a PCB **340**, a battery **350**, a second support member **360** (e.g., a rear case), an antenna **370**, and a rear plate **311**. The mobile electronic device **300** may omit at least one (e.g., the first support member **3211** or the second support member **360**) of the above components or may further include another component. Some components of the electronic device **300** may be the same as or similar to those of the mobile electronic device **300** shown in FIG. 3A or FIG. 3B, thus, descriptions thereof are omitted below.

(68) The first support member **3211** is disposed inside the mobile electronic device **300** and may be connected to, or integrated with, the lateral bezel structure **310**. The first support member **3211** may be formed of, for example, a metallic material and/or a non-metal (e.g., polymer) material. The first support member **3211** may be combined with the display **301** at one side thereof and also combined with the PCB **340** at the other side thereof. On the PCB **340**, a processor, a memory, and/or an interface may be mounted. The processor may include, for example, one or more of a CPU, an AP,

a GPU, an ISP, a sensor hub processor, or a CP.

(69) The memory may include, for example, volatile memory or non-volatile memory.

(70) The interface may include, for example, a high definition multimedia interface (HDMI), a USB interface, a secure digital (SD) card interface, and/or an audio interface. The interface may electrically or physically connect the mobile electronic device **300** with an external electronic device and may include a USB connector, an SD card/multimedia card (MMC) connector, or an audio connector.

(71) The battery **350** is a device for supplying power to at least one component of the mobile electronic device **300**, and may include, for example, a non-rechargeable primary battery, a rechargeable secondary battery, or a fuel cell. At least a part of the battery **350** may be disposed on substantially the same plane as the PCB **340**. The battery **350** may be integrally disposed within the mobile electronic device **300**, and may be detachably disposed from the mobile electronic device **300**.

(72) The antenna **370** may be disposed between the rear plate **311** and the battery **350**. The antenna **370** may include, for example, a near field communication (NFC) antenna, a wireless charging antenna, and/or a magnetic secure transmission (MST) antenna. The antenna **370** may perform short-range communication with an external device, or transmit and receive power required for charging wirelessly. An antenna structure may be formed by a part or combination of the lateral bezel structure **310** and/or the first support member **3211**.

(73) FIG. **4A(a)**, **(b)**, **(c)** are views illustrating a structure of, for example, a third antenna module described with reference to FIG. **2** according to various embodiments of the disclosure. FIG. **4A(a)** is a perspective view illustrating the third antenna module **246** viewed from one side and FIG. **4A(b)** is a perspective view illustrating the third antenna module **246** viewed from the other side. FIG. **4A(c)** is a cross-sectional view illustrating the third antenna module **246** taken along line X-X' of FIG. **4A(a)**.

(74) With reference to FIG. **4A(a)**, **(b)**, **(c)**, in one embodiment, the third antenna module **246** may include a printed circuit board **410**, an antenna array **430**, a RFIC **452**, and a PMIC **454** (see FIG. **4A(c)**). Alternatively, the third antenna module **246** may further include a shield member **490** (see FIG. **4A(c)**). In other embodiments, at least one of the above-described components may be omitted or at least two of the components may be integrally formed.

(75) The printed circuit board **410** may include a plurality of conductive layers and a plurality of non-conductive layers stacked alternately with the conductive layers. The printed circuit board **410** may provide electrical connections between the printed circuit board **410** and/or various electronic components disposed outside using wirings and conductive vias formed in the conductive layer.

(76) The antenna array **430** (e.g., **248** of FIG. **2**) may include a plurality of antenna elements **432**, **434**, **436**, or **438** disposed to form a directional beam. As illustrated in FIG. **4A(a)** and **(c)**, the antenna elements **432**, **434**, **436**, or **438** may be formed at a first surface of the printed circuit board **410**. According to another embodiment, the antenna array **430** may be formed inside the printed circuit board **410**. According to the embodiment, the antenna array **430** may include the same or a different shape or kind of a plurality of antenna arrays (e.g., dipole antenna array and/or patch antenna array).

(77) As shown in FIG. **4A(b)**, the RFIC **452** (e.g., the third RFIC **226** of FIG. **2**) may be disposed at another area (e.g., a second surface opposite to the first surface) of the printed circuit board **410** spaced apart from the antenna array. The RFIC **452** is configured to process signals of a selected frequency band transmitted/received through the antenna array **430**. According to one embodiment, upon transmission, the RFIC **452** may convert a baseband signal obtained from a communication processor (not shown) to an RF signal of a designated band. Upon reception, the RFIC **452** may convert an RF signal received through the antenna array **430** to a baseband signal and transfer the baseband signal to the communication processor.

(78) According to another embodiment, upon transmission, the RFIC **452** may up-convert an IF

signal (e.g., about 9 GHz to about 11 GHz) obtained from an intermediate frequency integrate circuit (IFIC) (e.g., **228** of FIG. 2) to an RF signal of a selected band. Upon reception, the RFIC **452** may down-convert the RF signal obtained through the antenna array **430**, convert the RF signal to an IF signal, and transfer the IF signal to the IFIC.

(79) The PMIC **454** may be disposed in another partial area (e.g., the second surface) of the printed circuit board **410** spaced apart from the antenna array **430**. The PMIC **454** may receive a voltage from a main PCB (not illustrated) to provide power necessary for various components (e.g., the RFIC **452**) on the antenna module.

(80) The shielding member **490** may be disposed at a portion (e.g., the second surface) of the printed circuit board **410** so as to electromagnetically shield at least one of the RFIC **452** or the PMIC **454**. According to one embodiment, the shield member **490** may include a shield can.

(81) Although not shown, in various embodiments, the third antenna module **246** may be electrically connected to another printed circuit board (e.g., main circuit board) through a module interface. The module interface may include a connecting member, for example, a coaxial cable connector, a board-to-board connector, an interposer, or a flexible printed circuit board (FPCB). The RFIC **452** and/or the PMIC **454** of the antenna module may be electrically connected to the printed circuit board through the connection member.

(82) FIG. 4B is a cross-sectional view illustrating the third antenna module **246** taken along line Y-Y' of FIG. 4A(a) according to various embodiments of the disclosure. The printed circuit board **410** of the illustrated embodiment may include an antenna layer **411** and a network layer **413**.

(83) With reference to FIG. 4B, the antenna layer **411** may include at least one dielectric layer **437-1**, and an antenna element **436** and/or a power feeding portion **425** formed on or inside an outer surface of a dielectric layer. The power feeding portion **425** may include a power feeding point **427** and/or a power feeding line **429**.

(84) The network layer **413** may include at least one dielectric layer **437-2**, at least one ground layer **433**, at least one conductive via **435**, a transmission line **423**, and/or a power feeding line **429** formed on or inside an outer surface of the dielectric layer.

(85) Further, in the illustrated embodiment, the RFIC **452** (e.g., the third RFIC **226** of FIG. 2) of FIG. 4A(c) may be electrically connected to the network layer **413** through, for example, first and second solder bumps **440-1** and **440-2**. In other embodiments, various connection structures (e.g., solder or a ball grid array (BGA)) instead of the solder bumps may be used. The RFIC **452** may be electrically connected to the antenna element **436** through the first solder bump **440-1**, the transmission line **423**, and the power feeding portion **425**. The RFIC **452** may also be electrically connected to the ground layer **433** through the second solder bump **440-2** and the conductive via **435**. Although not illustrated, the RFIC **452** may also be electrically connected to the above-described module interface through the power feeding line **429**.

(86) An electronic apparatus, which will be described below, may be an electronic apparatus **300** illustrated in FIGS. 3A to 3C. The electronic apparatus may include at least one of the constituent elements of the electronic apparatus **101** described with reference to FIG. 1.

(87) In the following description, an “electronic apparatus case” may be a lateral bezel structure **320** illustrated in FIGS. 3A and 3B or the lateral member **320** illustrated in FIG. 3C.

(88) In the following description, an “antenna member” may be the third antenna module **246** illustrated in FIGS. 2 and 4A. The term “antenna member” may refer to a constituent element that transmits or receives a communication signal through various communication methods including the communication method described with reference to FIG. 2.

(89) In the following description, the constituent elements identical or similar to the above-mentioned constituent elements will be assigned with the same reference numerals, and a repeated description thereof will be omitted.

(90) FIG. 5A is an exploded perspective view of an electronic apparatus case according to various embodiments disclosed in the present document. FIG. 5B is a front view of the electronic apparatus

case according to various embodiments disclosed in the present document.

(91) According to various embodiments, an electronic apparatus case **500** may support various electronic components and/or mechanisms included in the electronic apparatus. A part of the electronic apparatus case **500** may define an external appearance of the electronic apparatus. In the embodiment, the case **500** may include a plate **510**, a frame **520**, and an intermediate member **530**.

(92) According to various embodiments, the plate **510** may include a plurality of concave-convex portions **511**. Each one of the pair of concave-convex portions **511** may primarily fix various electronic components and/or mechanisms or define an area in which the electronic component and/or the mechanism is disposed. Therefore, a shape of each one of the concave-convex portions **511** may be variously changed depending on the number, shapes, and arrangements of the electronic components and/or mechanisms disposed in the electronic apparatus.

(93) In the embodiment, the plate **510** may be made of a metallic material. The plate **510** may be formed by various methods. The plate **510** may be made in various ways. For example, the plate **510** may be formed in various ways such as pressing, injection molding, casting including die casting, and cutting (e.g., CNC (computerized numerical control) milling). The above-mentioned forming methods are provided for illustrative purposes only. The plate **510** may be formed by various forming methods within a range that may be understood by those skilled in the art. In the embodiment, the plate **510** may be formed by casting such as die casting.

(94) According to various embodiments, the frame **520** may have a shape that surrounds the plate **510**. The plate **510** may be disposed in an opening **520A** formed in the frame **520**. In the embodiment, the frame **520** may be made of a metallic material. For example, the frame **520** may be made of a metallic material different from the metallic material of the plate **510** or made of a metallic material identical to the metallic material of the plate **510**. The frame **520** may be formed by various metal forming methods. For example, the frame **520** may be formed by processing an extrusion/rolling material through a machining method such as CNC.

(95) In the embodiment, the plate **510** and the frame **520** may be manufactured through different processing methods and coupled to each other through a joining method such as welding. In the embodiment, the plate **510** may be manufactured by performing die casting on an ingot. Because the ingot is relatively lower in material cost than the other material, it is possible to reduce a process unit price by using the ingot. Meanwhile, a material, which has excellent fluidity so that the material may smoothly flow in a mold, may be used to smoothly perform die casting. To improve fluidity of the ingot, it is possible to decrease a proportion of aluminum Al and increase a proportion of silicon Si. In this case, there may occur a problem in that it is difficult to apply an anodizing process to the plate **510**. The anodizing process is a process capable of providing luxuriousness to the material. In various embodiments disclosed in the present document, the frame **520** may define an external appearance of the electronic apparatus. The process unit price may be reduced when the frame **520** is manufactured by die casting using an ingot material in which a proportion of aluminum is decreased. However, as described above, there is a problem in that it is difficult to applying the anodizing process. Therefore, in consideration of the anodizing process, the frame **520**, which defines an external appearance of the electronic apparatus, may be manufactured through a process different from the process of manufacturing the plate **510**. For example, as described above, the frame **520** may be formed by processing an extrusion/rolling material, which has a large aluminum content, through a processing method such as CNC. The anodizing process may be easily applied to the frame **520** made of a material with a large aluminum content. Therefore, it is possible to provide luxuriousness to the frame **520** that defines an external appearance of the electronic apparatus.

(96) With reference to FIG. 5A, protruding portions **521** may be disposed along an inner periphery of the frame **520** and can protrude in a direction toward the inside of the frame **520**. A junction protrusion (e.g., a junction protrusion **811** in FIG. 8B) or a junction groove (e.g., a junction groove **812** in FIG. 8B) of a junction part (e.g., a junction part **810** in FIG. 8A) to be described below may

be formed on the protruding portion **521**. The protruding portion **521** may partially overlap a part of the plate **510** when the plate **510** is viewed in a direction (e.g., a +Z direction in FIG. 5A) toward the front surface of the electronic apparatus.

(97) According to various embodiments, the intermediate member **530** may be positioned between the frame **520** and the plate **510**. For example, the intermediate member **530** may be formed when a forming material injected between the frame **520** and the plate **510** is solidified. A melted forming material is injected between the frame **520** and the plate **510** and then solidified, such that the intermediate member **530** may be formed. The frame **520** and the plate **510**, which are joined to each other, may be inserted into a mold for forming the intermediate member **530**. The intermediate member **530** may be formed by injecting the forming material into the mold in the state in which the frame **520** and the plate **510** are inserted into the mold. The intermediate member **530** may be made of a metallic material or a synthetic resin material.

(98) According to various embodiments, an accommodation space **600** may be provided in a partial area between the frame **520** and the plate **510** and may accommodate an antenna member (e.g., an antenna member **700** in FIG. 6). The accommodation space **600** (see FIG. 6) may be a space defined by being surrounded by the frame **520**, the plate **510**, and the intermediate member **530**. For example, the accommodation space **600** may be formed by partially cutting the frame **520**, the plate **510**, and the intermediate member **530** or forming a space in which the accommodation space **600** is provided. In addition to the above-mentioned methods, the accommodation space **600** may be formed in various alternative ways. The partial area between the frame **520** and the plate **510** may be considered as a side surface **501** of the electronic apparatus. Therefore, the accommodation space **600** may be understood as a space formed in the side surface **501** of the electronic apparatus. The antenna member may be disposed in the accommodation space **600**. The antenna member may radiate a communication signal in the direction (e.g., a -X direction in FIG. 5B) toward the side surface of the electronic apparatus or receive a communication signal in the direction (e.g., the -X direction in FIG. 5B) toward the side surface of the electronic apparatus.

(99) As described above, the accommodation space **600** may mean a space formed when at least one of the frame **520**, the plate **510**, and the intermediate member **530** is machined or formed in a portion between the frame **520** and the plate **510**. In the following description, for convenience of description, the accommodation space **600** is indicated to a portion where the accommodation space **600** is to be formed.

(100) FIG. 6 is a perspective view illustrating the antenna member and the electronic apparatus case according to various embodiments disclosed in the present document. FIG. 7 is a perspective view of the antenna member according to various embodiments disclosed in the present document.

(101) The antenna member **700**, which will be described below, may be an antenna similar to the third antenna module **246** described with reference to FIGS. 4A and 4B.

(102) According to various embodiments, the antenna member **700** may include: a substrate **720** (e.g., the printed circuit board **410** in FIG. 4A); a radiation transmission/reception part (hereinafter referred to as “a radiation part”) **710** (e.g., the antenna array **430** in FIG. 4A); a radio frequency integrated circuit (RFIC) **730** (e.g., the RFIC **452** in FIG. 4A); a power management integrated circuit (PMIC) **740** (e.g., the PMIC **454** in FIG. 4A); and a shield member **750** (e.g., the shield member **490** FIG. 4A). The above-mentioned constituent elements of the antenna member **700** are provided for illustrative purposes only, and the antenna member **700** may not include some of the above-mentioned constituent elements.

(103) According to various embodiments, the radiation part **710** may be disposed on the substrate **720**. The radiation part **710** may transmit or receive a communication signal. The radiation part **710** may be provided as a plurality of radiation parts **710** that serves as an antenna array which is arranged on the substrate **720**. For example, as illustrated in FIG. 7, the antenna member **700** may include a first radiation part **711**, a second radiation part **712**, a third radiation part **713**, and a fourth radiation part **714**. The number of radiation parts **710** is provided for illustrative purposes only, and

the number of radiation parts **710** may be three or less or five or more, the radiation parts **710** may be aligned in a linear array or may be arranged in a different type of formation. The radiation part **710** may be disposed on a first surface **721** of the substrate **720** or disposed in the substrate **720**. In this case, the first surface **721** of the substrate **720** may be understood as a front surface of the antenna member **700**.

(104) In the embodiment, the radiation parts **710** may be provided on the first surface **721** of the substrate **720** and may be disposed at predetermined similar or varied intervals from one another. For example, the radiation parts **710** may be arranged at the same interval from one another in a longitudinal direction (e.g., a Y-axis direction in FIG. 7) of the substrate **720**. In another embodiment, the radiation parts **710** may be arranged at different intervals from one another in the longitudinal direction of the substrate **720**. FIGS. 6 and 7 illustrate that the plurality of radiation parts **710** has the same area and the same shape. However, as needed, the plurality of radiation parts **710** may have different areas and different shapes.

(105) According to various embodiments, the RFIC **730** may be disposed on the substrate **720**. For example, as illustrated in FIG. 7, the RFIC **730** may be disposed on a second surface **722** opposite to the first surface **721** on which the radiation part **710** is disposed. The RFIC **730** may be constituted to process a communication signal transmitted or received through the radiation part **710**. For example, the RFIC **730** may be electrically connected to a communication module (e.g., the communication module **180** in FIG. 1) of the electronic apparatus. The RFIC **730** may convert a signal processed by the communication module into a communication signal in a band designated by the radiation part **710** or convert a communication signal received by the radiation part **710** into a communication signal in a designated band and transmits the communication signal to the communication module. In the embodiment, the RFIC **730** may up-convert an intermediate frequency (IF) signal (e.g., about 9 GHz to about 11 GHz) to a communication signal in a designated band during an operation of transmitting the communication signal. The RFIC **730** may down convert a communication signal received through the radiation part **710** into an IF signal during an operation of receiving the communication signal.

(106) According to various embodiments, the PMIC **740** may be disposed on the substrate **720**. For example, as illustrated in FIG. 7, the PMIC **740** may be disposed on the second surface **722** opposite to the first surface **721** on which the radiation part **710** is disposed. The PMIC **740** may be electrically coupled to a main substrate (e.g., the printed circuit board **360** in FIG. 3C) of the electronic apparatus. The PMIC **740** may receive power transmitted from the main substrate and provide the respective constituent elements (e.g., the RFIC **730**) of the antenna member **700** with power required to operate the antenna member **700**.

(107) According to various embodiments, the shield member **750** may be disposed to electromagnetically shield at least one or both of the RFIC **730** and the PMIC **740**. For example, as illustrated in FIG. 7, the shield member **750** may be disposed to cover at least one or both of the RFIC **730** and the PMIC **740**. The shield member **750** may be made of a material capable of blocking electromagnetic interference (EMI). For example, the shield member **750** may include a shield can made of a metallic material or a film including a dielectric material and a conductor.

(108) According to various embodiments, the antenna member **700** may be electrically connected to the main substrate of the electronic apparatus through the connection member **750**. For example, the connection member **750** may include a coaxial cable connector, a board-to-board connector, an interposer, or a flexible printed circuit board (FPCB).

(109) According to various embodiments, an antenna bracket **790** may support the antenna member **700**. In the embodiment, the antenna bracket **790**, which supports the antenna member **700**, is fixed to the accommodation space **600** provided between the frame **520** and the plate **510**, such that the antenna member **700** may be seated in the accommodation space **600**. In the embodiment, the antenna bracket **790** may be coupled to the plate **510** by bolting.

(110) With reference to FIG. 6, in the embodiment, the antenna bracket **790** may be fixed to the

accommodation space **600** while supporting the antenna member **700** so that a side surface (e.g., a surface directed in a +Z direction in FIG. 7 or a surface directed in a -Z direction) of the antenna member **700** may be directed toward the front or rear surface of the electronic apparatus. In this case, the front surface of the electronic apparatus may refer to a surface on which a display module (e.g., the display **301** in FIG. 3C) of the electronic apparatus is disposed. The rear surface of the electronic apparatus may refer to a surface opposite to the front surface. When the side surface of the antenna member **700** is directed toward the rear or front surface of the electronic apparatus in the state in which the antenna member **700** is seated in the accommodation space **600**, the front surface (e.g., the surface directed in the -X direction in FIG. 7) of the antenna member **700** may be directed toward the side surface **501** of the electronic apparatus. The front surface of the antenna member **700** may be a surface corresponding to the first surface **721** of the substrate **720**. The radiation part **710** may be disposed on the first surface **721** of the substrate **720**. The radiation part **710** of the antenna member **700** may be directed toward the side surface **501** of the electronic apparatus.

(111) FIG. 8A is a view for explaining a coupling between the plate and the frame according to various embodiments disclosed in the present document. FIG. 8B is an enlarged view of area P illustrated in FIG. 8A. FIG. 9 is a view for explaining a positional relationship between the antenna member and a reinforcement junction part according to various embodiments disclosed in the present document.

(112) FIGS. 8A and 8B may be views illustrating the plate **510** and the frame **520** in an intermediate step of a process of manufacturing the electronic apparatus case. This state may be a step before the intermediate member **530** is formed.

(113) According to various embodiments, the junction part **810** may mean a portion where the plate **510** and the frame **520** are coupled. A plurality of junction parts **810** may be present along an outer periphery of the plate **510** and an inner periphery of the junction part **810**.

(114) According to various embodiments, the junction part **810** may include a junction protrusion **811** and a junction groove **812**. The junction protrusion **811** and the junction groove **812** may be constitutions corresponding to each other. The junction protrusion **811** may be formed to be inserted into the junction groove **812**. The junction protrusion **811** may be inserted into the junction groove **812**, such that a relative position between the plate **510** and the frame **520** may be determined. The plate **510** and the frame **520** may be joined to each other at a designed position by joining the junction protrusion **811** to the junction groove **812** in the state in which the relative position between the plate **510** and the frame **520** is determined.

(115) In the embodiment, the junction protrusion **811** may be provided as a plurality of junction protrusions **811** provided along the outer periphery of the plate **510**, and the junction groove **812** may be provided as a plurality of junction grooves **812** provided along the inner periphery of the frame **520**. Although not illustrated in the drawings, because the junction groove **812** and the junction protrusion **811** are constitutions corresponding to each other, the plurality of junction protrusions **811** may be provided along the inner periphery of the frame **520**, and the plurality of junction grooves **812** may be provided along the outer periphery of the plate **510** and disposed at positions corresponding to the junction protrusions **811**. In some instances, some of the plurality of junction grooves **812** may be formed in the frame **520**, and the remaining junction grooves **812** may be formed in the plate **510**. In this case, some of the junction protrusions **811** may be formed on the plate **510** and disposed at the positions corresponding to the junction grooves **812**, and the remaining junction protrusions **811** may be formed on the frame **520**.

(116) With reference to FIG. 8B, the junction groove **812** may be formed in the protruding portion **521** protruding from the frame **520** in the direction (e.g., the +X direction in FIG. 8B) toward the plate **510**. According to various embodiments, the junction protrusion **811** and the junction groove **812** may be integrated with the plate **510** and the frame **520**, respectively. For example, the junction protrusion **811** may be formed together with the plate **510** during the process of forming the plate

510. The junction groove **812** may be formed together with the frame **520** during the process of forming the frame **520**.

(117) According to various embodiments, the junction protrusion **811** is inserted into the junction groove **812**, which corresponds in position to the junction protrusion **811**, such that the relative position between the frame **520** and the plate **510** may be primarily fixed. The frame **520** and the plate **510** may be coupled to each other by maintaining, by various methods, the state in which the junction protrusion **811** is inserted into the junction groove **812**. For example, the junction protrusion **811** may be coupled to the junction groove **812** by a method such as bonding using a bonding agent or welding. In case the junction protrusion **811** is coupled to the junction groove **812** by welding, a surface of the junction protrusion **811** and a surface of the junction groove **812**, which adjoin each other, may be welded, or a portion of the junction protrusion **811** and a portion of the junction groove **812**, which adjoin each other, may be partially welded by spot welding. The junction protrusion **811** and the junction groove **812** may be joined to each other by various welding methods. For example, welding methods, such as arc welding, gas welding, laser welding, and electron beam welding, may be used. In addition to the above-mentioned welding methods, the junction protrusion **811** and the junction groove **812** may be coupled to each other in various ways.

(118) According to various embodiments, as illustrated in FIG. **8B**, in a part of the junction part **810**, a diameter of the junction protrusion **811** may be smaller than a diameter of the junction groove **812**. A separation space **813** may be formed between the junction protrusion **811** and the junction groove **812** in the state in which the junction protrusion **811** is inserted into the junction groove **812**. The plate **510** and the frame **520** may be inserted into the mold in the state in which the plate **510** and the frame **520** are coupled to each other by means of the junction part **810**. In this state, the intermediate member **530** may be formed by an injection of the forming material. The forming material may smoothly reach a portion, into which the forming material needs to be injected, while passing through the separation space **813** between the junction protrusion **811** and the junction groove **812**.

(119) According to various embodiments, the junction parts **810** may include a reinforcement junction part **820**. Like the junction part **810**, the reinforcement junction part **820** may include a junction protrusion **821** and a junction groove **822**. The reinforcement junction part **820** may be understood as the junction part **810** disposed adjacent to the accommodation space **600** in which the antenna member **700** is disposed. The reinforcement junction part **820** may be disposed between the adjacent radiation parts **710** among the plurality of radiation parts **710** included in the antenna member **700**. The reinforcement junction part **820** may be disposed adjacent to the accommodation space **600** and reinforce a coupling force between the plate **510** and the frame **520** at the portion of the accommodation space **600**.

(120) For example, in case the junction protrusion **811** and the junction groove **812** are made of a metallic material, it may be difficult for the communication signal to pass through the junction part **810**. The junction protrusion **811** and the junction groove **812**, which face the radiation part **710** of the antenna member **700**, may act as obstacles when the antenna member **700** disposed in the accommodation space **600** receives or transmits the communication signal. In particular, because straightness of the communication signal is high, the influence of the junction protrusion **811** and the junction groove **812** may become greater when the antenna member **700** transmits or receives a high-frequency communication signal (e.g., an mmWave). In case the junction protrusion **811** and the junction groove **812** are not joined to each other at the portion adjacent to the portion where the antenna member **700** is disposed (e.g., the accommodation space **600**) to prevent the influence, the portion where the accommodation space **600** is positioned becomes a weak point of the frame **520** and the plate **510**, which may cause a decrease in strength of the case **500**.

(121) With reference to FIG. **8B**, the reinforcement junction part **820** according to various embodiments disclosed in the present document may be positioned at the portion adjacent to the accommodation space **600**. The frame **520** and the plate **510** are joined to each other at the portion

adjacent to the accommodation space **600** by the reinforcement junction part **820**, which makes it possible to solve the above-mentioned problem. Further, the reinforcement junction part **820** is disposed while avoiding the radiation part **710** of the antenna member **700**, which makes it possible to reduce an influence on the communication signal transmitted or received through the antenna member **700**. As illustrated in FIG. **8B**, because the reinforcement junction part **820** is disposed between the radiation parts **710**, it is possible to reduce the extent to which the communication signal, which is transmitted from the radiation part **710** or received through the radiation part **710**, interferes with the reinforcement junction part **820**. For example, in case the reinforcement junction part **820** is made of a metallic material, the reinforcement junction part **820** may act as an obstacle to the transmission or reception of the communication signal. Because the reinforcement junction part **820** is disposed between the radiation parts **710**, it is possible to reduce the communication interference caused by the reinforcement junction part **820**.

(122) FIG. **9** is a view for explaining a positional relationship between the antenna member **700** and the reinforcement junction part **820** when the case **500** is viewed from the side surface in a state in which the frame **520** is removed. For example, in case the radiation parts **710** include the first radiation part **711**, the second radiation part **712**, the third radiation part **713**, and the fourth radiation part **714**, the junction protrusion **821** of the reinforcement junction part **820** may be disposed between the second radiation part **712** and the third radiation part **713**, as illustrated in FIGS. **8B** and **9**. In another embodiment, the reinforcement junction part **820** may be disposed between the first radiation part **711** and the second radiation part **712** or between the third radiation part **713** and the fourth radiation part **714**. In this case, the configuration in which the reinforcement junction part **820** is disposed between the radiation parts **710** may mean that the reinforcement junction part **820** is positioned between the radiation parts **710** when the electronic apparatus is viewed from the side surface. In the embodiment, a distance **L1** between the junction protrusion **821** of the reinforcement junction part **820** and the second radiation part **712** may be substantially equal to a distance **L2** between the junction protrusion **821** and the third radiation part **713**.

(123) According to various embodiments, the reinforcement junction part **820** may be disposed between the first radiation part **711** and the second radiation part **712** or disposed between the third radiation part **713** and the fourth radiation part **714**. In some instances, the plurality of reinforcement junction parts **820** may be disposed in at least one of the portions between the first radiation part **711** and the second radiation part **712**, between the second radiation part **712** and the third radiation part **713**, and between the third radiation part **713** and the fourth radiation part **714**.

(124) FIG. **10A** is a flowchart illustrating a method of manufacturing the electronic apparatus case according to various embodiments disclosed in the present document. FIG. **10B(a)**, **(b)**, **(c)**, **(d)** is a view illustrating a state of the electronic apparatus case manufactured according to various embodiments disclosed in the present document. FIG. **10C** is a view illustrating states of the accommodation space and a peripheral portion of the electronic apparatus case manufactured according to various embodiments disclosed in the present document.

(125) According to various embodiments, the electronic apparatus case **500** may be manufactured in accordance with the flowchart illustrated in FIG. **10A**. However, the flowchart illustrated in FIG. **10A** does not limit the method of or the order of the process of manufacturing the electronic apparatus case **500**. The flowchart illustrated in FIG. **10A** is provided for illustrative purposes only, and the electronic apparatus case **500** may be manufactured in accordance with various ways and orders within a range that may be understood by those skilled in the art. The method of manufacturing the case **500**, which will be described below, is provided to exemplarily explain one of the methods of manufacturing the above-mentioned case **500**. The above-mentioned case **500** is not limited by the case **500** according to these methods.

(126) First, the frame **520** and the plate **510** may be formed dependently or independently (**1010**). For example, the plate **510** and the frame **520** may be formed by various processing methods (e.g., plastic processing or cutting). According to various embodiments, a fixing part **529** capable of

fixing the frame **520** to a processing apparatus may be formed during the process of forming the frame **520**. For example, as illustrated in FIG. **10B(a)**, (b), (c), (d), the fixing part **529** may be a portion formed on the outer periphery of the frame **520**.

(127) Next, the frame **520** and the plate **510** may be temporarily assembled (**1020**). In the embodiment, the plate **510** and the frame **520** may be temporarily assembled. FIG. **10B(b)** is a view illustrating a state in which the plate **510** and the frame **520** are temporarily assembled. When the frame **520** is fixed to the processing apparatus by the fixing part **529**, the plate **510** may be disposed in the frame **520**. For example, as illustrated in FIG. **10C(b)**, the plate **510** and the frame **520** may be temporarily assembled by inserting the junction protrusion **811**, which is formed along the outer periphery of the plate **510**, into the junction groove **812** formed along the inner periphery of the frame **520**.

(128) According to various embodiments, the frame **520** and the plate **510** may be joined and coupled to each other (**1030**). For example, the junction protrusion **811** inserted into the junction groove **812** may be welded to the junction groove **812**. When the junction groove **812** and the junction protrusion **811** are welded, the frame **520** and the plate **510** may be coupled. In the embodiment, a diameter of the junction protrusion **811** may be smaller than a diameter of the junction groove **812**. In the state in which the junction protrusion **811** is inserted into the junction groove **812**, the separation space may be formed between the junction protrusion **811** and the junction groove **812**.

(129) According to various embodiments, the intermediate member **530** may be formed between the frame **520** and the plate **510** (**1040**). For example, the frame **520** and the plate **510** are inserted into the mold, and the melted forming material may be injected into the mold. The forming material may be injected between the frame **520** and the plate **510**. The space between the frame **520** and the plate **510** may be smoothly filled with the melted forming material through the separation space (e.g., the separation space **813** in FIG. **8B**) between the junction protrusion **811** and the junction groove **812**. The intermediate member **530** may be formed when the forming material is solidified. FIG. **10B(c)** is a view illustrating a state in which the intermediate member **530** is formed.

(130) Next, cutting processing may be performed on the frame **520**, the plate **510**, and the intermediate member **530** (**1050**). In the cutting processing step, the fixing part **529** for fixing the frame **520** to the processing apparatus may be cut, and the outer surface of the frame **520** may be polished. FIG. **10B(d)** is a view of the case **500** on which the cutting processing has been completely performed.

(131) According to various embodiments, the accommodation space **600**, in which the antenna member **700** is to be disposed, may be formed by cutting a part of the intermediate member **530** disposed between the frame **520** and the plate **510**. For example, as illustrated in FIG. **10C(c)**, the accommodation space **600** may be formed by cutting processing. In the embodiment, a bolt coupling groove **512** may be formed to fix the antenna bracket **790**, which supports the antenna member **700**, to the accommodation space **600**. In this case, as illustrated in FIG. **10C(d)**, the antenna bracket **790** may be coupled to the plate **510** by bolting by means of the bolt coupling groove **512**.

(132) According to various embodiments, the plurality of radiation parts **710** of the antenna member **700**, which is seated in the accommodation space **600**, may be disposed so as not to overlap the reinforcement junction part **820** disposed adjacent to the accommodation space **600** when the case **500** is viewed from the side surface. The reinforcement junction part **820** may be disposed between the adjacent radiation parts. For example, as illustrated in FIG. **10C(d)**, the reinforcement junction part **820** may be disposed between the second radiation part **712** and the third radiation part **713** that are disposed adjacent to each other. The reinforcement junction part **820** may reinforce the coupling force between the plate **510** and the frame **520** at the portion of the accommodation space **600**. In addition, the reinforcement junction part **820** may be disposed between the radiation parts **710** and reduce interference with the communication signal transmitted

or received through the radiation part **710**.

(133) FIG. **11(a)**, **(b)** is a cross-sectional view taken by cutting the electronic apparatus case illustrated in FIG. **10C(a)**, **(b)**, **(c)**, **(d)**. FIG. **11(a)** is a cross-sectional view taken along line A-A in FIG. **10C(d)**, and FIG. **11(b)** is a cross-sectional view taken along line B-B in FIG. **10C(d)**.

(134) According to various embodiments, because the reinforcement junction part **820** is disposed between the radiation parts **710**, the radiation part **710** of the antenna member **700** is not illustrated in FIG. **11(a)** that is a cross-sectional view taken by cutting the electronic apparatus case to include the reinforcement junction part **820**.

(135) With reference to FIG. **11(b)**, a positional relationship between the frame **520** and the radiation part **710** may be adjusted so that the radiation part **710** of the antenna member **700** disposed in the accommodation space **600** does not face the frame **520**. For example, the shape of the frame **520** and the arrangement position of the radiation part **710** may be adjusted so that the frame **520** and the radiation part **710** do not overlap each other when the electronic apparatus is viewed from the side surface. In case the frame **520** is made of a metallic material, the frame **520** may interfere with the communication signal transmitted from or received by the radiation part **710**. Therefore, the shape of the frame **520** may be determined so that the frame **520** is not positioned in a radial direction of the radiation part **710**.

(136) With reference to FIG. **11(b)**, a height **H1** of the frame **520** may be smaller than a height **H2** of the radiation part **710**. In this case, the height **H1** of the frame **520** may mean a length from the portion adjacent to the accommodation space **600** to the portion maximally protruding from the rear surface of the electronic apparatus. The height of the radiation part **710** may mean a distance by which the radiation part **710** is spaced apart from the rear surface of the electronic apparatus.

(137) The electronic apparatus (e.g., the electronic apparatus **101** in FIG. **1** or the electronic apparatus **300** in FIG. **3A**) according to various embodiments disclosed in the present document may include: the frame (e.g., the frame **520** in FIG. **5A**); the plate (e.g., the plate **510** in FIG. **5A**) disposed in the frame; the antenna member (e.g., the antenna member **700** in FIG. **7**) including the plurality of radiation parts (e.g., the radiation parts **710** in FIG. **7**) configured to transmit or receive the RF signal; the accommodation space (e.g., the accommodation space **600** in FIG. **6**) provided between the frame and the plate so that the antenna member is disposed between the frame and the plate; and the plurality of junction parts (e.g., the junction parts **810** in FIG. **8A**) that is parts by which the frame and the plate are joined. The reinforcement junction part (e.g., the reinforcement junction part **820** in FIG. **8A**) included in the plurality of junction parts may be positioned between the adjacent radiation parts among the plurality of radiation parts of the antenna member.

(138) In addition, the electronic apparatus may further include the display module (e.g., the display module **160** in FIG. **1** and the display **301** in FIG. **3C**) disposed on the front surface of the electronic apparatus. The antenna member may be disposed in the accommodation space so that the plurality of radiation parts is directed toward the side surface of the electronic apparatus.

(139) In addition, the antenna member may be disposed in the accommodation space so that the first surface of the substrate (e.g., the substrate **720** in FIG. **7**) included in the antenna member is directed toward the side surface of the electronic apparatus. The plurality of radiation parts of the antenna member may be provided on the first surface (e.g., the first surface **721** in FIG. **7**) of the substrate and disposed at predetermined intervals.

(140) In addition, the antenna member may include: the radio frequency integrated circuit (RFIC) (e.g., the RFIC **730** in FIG. **7**) disposed on the second surface (e.g., the second surface **722** in FIG. **7**) opposite to the first surface of the substrate and electrically connected to the radiation part; and the shield member (e.g., the shield member **750** in FIG. **7**) disposed on the second surface of the substrate and configured to shield the RFIC.

(141) In addition, the plurality of radiation parts of the antenna member may be arranged in the longitudinal direction of the substrate.

(142) In addition, the plurality of radiation parts of the antenna member may include the first

radiation part (e.g., the first radiation part **711** in FIG. 7), the second radiation part (e.g., the second radiation part **712** in FIG. 7), the third radiation part (e.g., the third radiation part **713** in FIG. 7), and the fourth radiation part (e.g., the fourth radiation part **714** in FIG. 7). The reinforcement junction part may be positioned between the second radiation part and the third radiation part when the electronic apparatus is viewed from the side surface.

(143) In addition, the distance between the reinforcement junction part and the second radiation part may be substantially equal to the distance between the reinforcement junction part and the third radiation part when the electronic apparatus is viewed from the side surface.

(144) In addition, the frame and the plate may be made of a metallic material, and the plurality of junction parts may be parts where the frame and the plate are coupled to each other by welding.

(145) In addition, at least one of the plurality of junction parts may include: the junction protrusion (e.g., the junction protrusion **811** in FIG. 8B) formed on one of the frame and the plate; and the junction groove (e.g., the junction groove **812** in FIG. 8B) formed in the other of the frame and the plate and provided at a position corresponding to the junction protrusion so that the junction protrusion may be inserted into the junction groove.

(146) In addition, the electronic apparatus may further include the intermediate member (e.g., the intermediate member **530** in FIG. 5A) provided between the frame and the plate.

(147) In addition, the intermediate member may be formed by injecting a forming material between the frame and the plate.

(148) In addition, the diameter of the junction protrusion of at least one of the plurality of junction parts may be smaller than the diameter of the junction groove so that the forming material passes.

(149) The electronic apparatus case (e.g., the case **500** in FIG. 5A) according to various embodiments disclosed in the present document may include: the frame (e.g., the frame **520** in FIG. 5A); the plate (e.g., the plate **510** in FIG. 5A) disposed in the frame; the plurality of junction parts (e.g., the junction part **810** in FIG. 8A) that is parts by which the frame and the plate are joined; and the intermediate member (e.g., the intermediate member **530** in FIG. 5A) formed between the frame and the plate. The reinforcement junction part (e.g., the reinforcement junction part **820** in FIG. 8A) included in the plurality of junction parts may be positioned between the adjacent radiation parts among the plurality of radiation parts (e.g., the radiation parts **710** in FIG. 7) included in the antenna member (e.g., the antenna member **700** in FIG. 7) disposed in the accommodation space (e.g., the accommodation space **600** in FIG. 6) provided between the frame and the plate.

(150) In addition, at least one of the plurality of junction parts may include: the junction protrusion (e.g., the junction protrusion **811** in FIG. 8B) formed on one of the frame and the plate; and the junction groove (e.g., the junction groove **812** in FIG. 8B) formed in the other of the frame and the plate and provided at a position corresponding to the junction protrusion so that the junction protrusion may be inserted into the junction groove.

(151) In addition, the frame and the plate may be made of a metallic material, and the plurality of junction parts may be parts where the frame and the plate are coupled to each other by welding.

(152) In addition, the reinforcement junction part may be positioned between the adjacent radiation parts among the plurality of radiation parts included in the antenna member when the electronic apparatus case is viewed from the side surface.

(153) The method of manufacturing the electronic apparatus case (e.g., the case **500** in FIG. 5A) according to various embodiments disclosed in the present document may include: forming the plate (e.g., the plate **510** in FIG. 5A) and the frame (e.g., the frame **520** in FIG. 5A) (e.g., **1010** in FIG. 10A); temporarily assembling the frame to the plate so that the junction protrusion (e.g., the junction protrusion **811** in FIG. 8B) formed on one of the frame and the plate is inserted into the junction groove (e.g., the junction groove **812** in FIG. 8B) formed on the other of the frame and the plate (e.g., **1020** in FIG. 10A); joining the junction protrusion and the junction groove (e.g., **1030** in FIG. 10A); forming the intermediate member (e.g., the intermediate member **530** in FIG. 5A) in the

space between the frame and the plate (e.g., **1040** in FIG. **10A**); and cutting at least one of the frame, the plate, and the intermediate member (e.g., **1050** in FIG. **10A**). The cutting of the at least one of the frame, the plate, and the intermediate member may include forming the accommodation space (e.g., the accommodation space **600** in FIG. **6**) in which the antenna member (e.g., the antenna member **700** in FIG. **7**) is disposed between the frame and the plate. The reinforcement junction protrusion (e.g., the junction protrusion **821** in FIG. **8B**) included in the junction protrusion and the reinforcement junction groove (e.g., the junction protrusion **822** in FIG. **8B**) included in the junction groove may be positioned between the adjacent radiation parts among the plurality of radiation parts (e.g., the radiation parts **710** in FIG. **7**) included in the antenna member disposed in the accommodation space.

(154) In addition, the reinforcement junction protrusion and the reinforcement junction groove may be positioned between the adjacent radiation parts among the plurality of radiation parts included in the antenna member when the case is viewed from the side surface.

(155) In addition, the junction protrusion and the junction groove may be joined by welding.

(156) In addition, the forming of the intermediate member may include forming the intermediate member by injection molding by inserting the plate and the frame into the mold and injecting a forming material into the mold.

(157) Further, the embodiments disclosed in the present document disclosed in the present specification and illustrated in the drawings are provided as particular examples for easily explaining the technical contents according to the embodiment disclosed in the present document and helping understand the embodiment disclosed in the present document, but not intended to limit the scope of the embodiment disclosed in the present document. Accordingly, the scope of the various embodiments disclosed in the present document should be interpreted as including all alterations or modifications derived from the technical spirit of the various embodiments disclosed in the present document in addition to the embodiments disclosed herein.

Claims

1. An electronic apparatus comprising: a frame; a plate disposed in the frame; an antenna member disposed in an accommodation space provided between the frame and the plate, the antenna member comprising a plurality of radiation parts configured to transmit or receive an RF signal; and a plurality of junction parts by which the frame and the plate are joined, the plurality of junction parts comprising a reinforcement junction part positioned between the plurality of radiation parts, wherein the reinforcement junction part comprises: a reinforcement junction protrusion formed on one of the frame and the plate; and a reinforcement junction groove formed in the other of the frame and the plate and provided at a position corresponding to the reinforcement junction protrusion so that the reinforcement junction protrusion is inserted into the reinforcement junction groove.
2. The electronic apparatus of claim 1, further comprising: a display module disposed on a front surface of the electronic apparatus, wherein the antenna member is disposed in the accommodation space so that the plurality of radiation parts is directed toward a side surface of the electronic apparatus.
3. The electronic apparatus of claim 1, wherein the antenna member is disposed in the accommodation space so that a first surface of a substrate included in the antenna member is directed toward a side surface of the electronic apparatus, and wherein the plurality of radiation parts of the antenna member is provided on the first surface of the substrate and disposed at predetermined intervals.
4. The electronic apparatus of claim 3, wherein the antenna member comprises: a radio frequency integrated circuit (RFIC) electrically connected to the plurality of radiation parts and disposed on a second surface opposite to the first surface of the substrate; and a shield member disposed on the

second surface of the substrate and configured to shield the RFIC.

5. The electronic apparatus of claim 3, wherein the plurality of radiation parts of the antenna member is arranged in a longitudinal direction of the substrate.

6. The electronic apparatus of claim 1, wherein the plurality of radiation parts of the antenna member comprises a first radiation part, a second radiation part, a third radiation part, and a fourth radiation part, and wherein the reinforcement junction part is positioned between the second radiation part and the third radiation part when the electronic apparatus is viewed from a side surface.

7. The electronic apparatus of claim 6, wherein a distance between the reinforcement junction part and the second radiation part is substantially equal to a distance between the reinforcement junction part and the third radiation part when the electronic apparatus is viewed from the side surface.

8. The electronic apparatus of claim 1, wherein the frame and the plate are each made of a metallic material, and wherein the plurality of junction parts is parts by which the frame and the plate are coupled by welding.

9. The electronic apparatus of claim 1, wherein at least one of the plurality of junction parts comprises: a junction protrusion formed on one of the frame and the plate; and a junction groove formed in the other of the frame and the plate and provided at a position corresponding to the junction protrusion so that the junction protrusion is inserted into the junction groove.

10. The electronic apparatus of claim 9, further comprising: an intermediate member formed between the frame and the plate.

11. The electronic apparatus of claim 10, wherein the intermediate member is formed by injecting a forming material between the frame and the plate.

12. The electronic apparatus of claim 11, wherein a diameter of the junction protrusion of at least one of the plurality of junction parts is smaller than a diameter of the junction groove so that the forming material passes.

13. An electronic apparatus case comprising: a frame formed of a metal material; a plate disposed in the frame and formed of a metal material; a plurality of junction parts that comprises parts by which the frame and the plate are joined through welding; and an intermediate member formed between the frame and the plate, wherein a reinforcement junction part included in the plurality of junction parts is positioned between adjacent radiation parts among a plurality of radiation parts included in an antenna member disposed in an accommodation space provided between the frame and the plate, and wherein at least one of the plurality of junction parts comprises: a junction protrusion formed on one of the frame and the plate; and a junction groove formed in the other of the frame and the plate and provided at a position corresponding to the junction protrusion so that the junction protrusion is inserted into the junction groove.

14. A method of manufacturing an electronic apparatus case, the method comprising: forming a plate and a frame; temporarily assembling the frame to the plate so that a junction protrusion formed on one of the frame and the plate is inserted into a junction groove formed in the other of the frame and the plate; joining the junction protrusion and the junction groove; forming an intermediate member in a space between the frame and the plate; and machining at least one of the frame, the plate, and the intermediate member, wherein the cutting of the at least one of the frame, the plate, and the intermediate member comprises forming an accommodation space in which an antenna member is disposed between the frame and the plate, and wherein a reinforcement junction protrusion included in the junction protrusion and a reinforcement junction groove included in the junction groove is positioned between a plurality of radiation parts included in the antenna member disposed in the accommodation space, and wherein the reinforcement junction protrusion and the reinforcement junction groove are positioned between the plurality of radiation parts included in the antenna member when the case is viewed from a side surface.

15. The method of claim 14, wherein the reinforcement junction protrusion and the reinforcement

junction groove are positioned between the adjacent radiation parts among the plurality of radiation parts included in the antenna member when the case is viewed from a side surface.
